Kefan Xu

UX RESEARCH / APPLIED HCI

PhD expected Spring 2027

Atlanta, GA | kefanxu@gatech.edu | Portfolio | Research | LinkedIn | GitHub

UX researcher and fifth-year CS HCI PhD candidate who takes health technologies from formative research through interaction design, software development, deployment, and longitudinal field evaluation. Design and build mobile and AI-assisted clinical tools; conduct mixed-methods user studies spanning **28–42 days**. First-author publications at **CHI and CSCW**.

RESEARCH EXPERIENCE

Georgia Institute of Technology

2022 – Present

Graduate Research Assistant | Ubicomp Health and Wellness Lab

CareWork AI-assisted clinical workflows

- Designed and developed an LLM-assisted clinician dashboard informed by **six formative interviews**; implemented editable suggestions and clinician approval across care-management tasks.
- Evaluated with **11 healthcare professionals** using simulated care scenarios and think-aloud interviews; identified requirements for verifiability, personalization, and workflow integration.

EcoCare AI support for navigating changes in chronic care; evaluation pending

- Designed and developed an **AI-assisted web visualization** that maps connections among patients, caregivers, clinicians, and care elements. Integrated an **AI assistant** to explain how life changes disrupt care and help users explore coping strategies.

Moodloop Mental-state reporting in everyday life

- Designed and developed the mobile app from Figma flows; built a companion app for researchers to tailor interviews using reporting histories.
- Deployed a **42-day field study with 15 participants**; conducted interviews and contributed to thematic analysis of **45 interviews** on scale choices and contextual annotations.

Trackya Physical-activity tracking and reflection

- Designed and developed a mobile tracker and activity dashboard; completed a **28-day field study with 18 participants** (**54 interviews**). Currently analyzing the interviews and writing the paper on everyday physical-activity reflection.

DUCSS Diabetes self-management

- Interviewed **22 patients** and used reflexive thematic analysis to characterize care needs; developed a diabetic foot ulcer monitoring system spanning SMS, a mobile app, and a clinician dashboard.

Caregiving over time Online health communities

- Analyzed **427 posts and 888 comments** from online health communities; developed design recommendations addressing conflicts and changing support needs (first-author CSCW 2025).

PECSS Clinical tools for PTSD therapy

- Collaborated with Emory clinicians to evaluate and implement sensing systems supporting PTSD therapy.

University of Michigan

2019 – 2022

Graduate Research Assistant | School of Information

Planneregy Activity planning; continued at Georgia Tech

- Designed and developed a **React Native** physical activity planning app and deployed through TestFlight.
- Conducted a **42-day field study with 16 participants**, combining interviews, weekly questionnaires, and **434 activity plans** (first-author CHI 2024).

RESEARCH EXPERIENCE (CONTINUED)

University of Michigan (continued) 2019 – 2022

Graduate Research Assistant | School of Information

Physicify Learning from activity-planning histories

- Designed, developed, and deployed a physical activity tracking app in a **28-day field study with 17 participants**; analyzed interviews and **248 activity plans** to understand how historical records support planning (first-author CHI 2022).

Pace to Plan Activity pacing for chronic fatigue

- Implemented a React Native app for older adults, supporting hybrid experience sampling and low-burden reporting.

Michigan Medicine, University of Michigan 2021 – 2022

Research Intern

Taperology Clinical decision support

- Designed and deployed a web benzodiazepine taper calculator; translated clinical partners' workflow needs into a decision-support interface.

METHODS & TECHNICAL SKILLS

Research Semi-structured interviews, contextual inquiry, think-aloud usability evaluation, surveys, diary and experience sampling studies, co-design, longitudinal field studies.

Analysis Reflexive thematic analysis, qualitative content analysis, affinity mapping, research synthesis, Dovetail.

Design Interaction design, wireframes, mobile and web prototyping, Figma.

Development React Native, Swift, Expo, Python, JavaScript, HTML/CSS, Java, C++, Git.

EDUCATION

PhD, Computer Science (HCI Track) Georgia Tech — Expected Spring 2027

MS, Information Science University of Michigan — 2021

BS, Data Science; BS, Interactive Media Arts NYU — 2019

SELECTED PUBLICATIONS

CSCW 2025 Caregivers' sense-making in online health communities. First author.

CHI 2024 Reflective iteration in physical-activity planning. First author.

CHI 2022 Historical records in activity-plan creation and execution. First author.

DIS 2025 Explainable AI for daily scenarios: end-user perspectives. Coauthor.

CHI 2026 Clinical decision support for outpatient veteran PTSD care. Coauthor.

Topics shown above; full titles and additional publications: kefanxu.com

LEADERSHIP & RECOGNITION

Teaching (2025–2026) Designed and developed a 16-week mobile well-being course at Georgia Tech and taught it for three semesters.

Mentoring (2022–Present) Mentored four graduate researchers on development and study protocols.

Recognition OMSCS Pre-Doc Fellowship (2025); Google Health Equity Research Initiative (2022).